

An Introduction to E-Commerce

with a Focus on Machine Learning, Deep Learning, and Artificial Intelligence

Lecture 09: Pricing and Promotions with ML + Optimization

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Session plan (90 minutes)

- 1 Why pricing is a core AI problem in commerce
- 2 Pricing objectives
- 3 Price, promotion, and discount structure
- 4 Demand forecasting
- 5 Price elasticity
- 6 Hands-on lab: forecast plus constrained optimization

Learning objectives

By the end of this lecture, you should be able to:

- Explain why pricing in e-commerce is both a prediction problem and an optimization problem.
- Distinguish demand forecasting, price elasticity estimation, and promotion uplift modeling.
- Analyze how business constraints such as inventory, margins, contracts, and competition shape pricing decisions.
- Design a pricing workflow that combines forecasting, optimization, and experimentation rather than treating price as a static input.

Why pricing is a core AI problem in commerce

- demand estimation,
- elasticity understanding,
- business-rule enforcement,
- optimization under constraints,
- and experimentation to measure causal impact.

- **Common objectives:** **Revenue maximization:** increase gross sales volume.; **Margin or contribution optimization:** emphasize profitability rather than top-line revenue.
- **Why objective clarity matters:** clearing inventory,; preserving premium brand position,

- **Promotions as structured interventions:** percentage discount,; fixed-amount coupon,

- **What is being forecast:** unit demand by SKU and day;; category demand,
- **Inputs to demand models:** historical sales;; current and historical prices,
- **Why forecasting is not enough:** Forecasting predicts what may happen.

- **Intuition:** If a small increase in price causes a large reduction in demand, the product is relatively price-sensitive.
- **Why elasticity is difficult to estimate:** In live commerce systems, observed prices are not randomly assigned.
- **Practical considerations:** category,; season,

Promotions and uplift

- **Observed uplift versus causal uplift:** genuine causal impact from the promotion,; seasonal demand shifts,
- **Why uplift matters:** Promotions can be expensive.

Optimization under business constraints

- **Typical constraints:** minimum margin thresholds,; inventory availability,
- **Why unconstrained optimization is unrealistic:** violates margin policy,; creates stockouts,

- **When dynamic pricing can help:** demand fluctuates quickly,; inventory is perishable or time-sensitive,
- **When dynamic pricing can hurt:** customer trust may decline if prices feel unstable or unfair,; operations may struggle with frequent updates,

- **Competitive signals:** competitor list price,; promotion intensity,
- **Strategic caution:** A competitor may use a loss leader or temporary campaign that should not be copied directly.

B2C versus B2B pricing

- **B2C:** high-volume SKU-level decisions,; campaign responsiveness,
- **B2B:** negotiated price lists,; account-specific discounts,

Feature design for pricing models

- **Feature governance:** Pricing is high impact, so feature governance matters.

Experimentation in pricing

- **Why pricing experiments are sensitive:** affect revenue directly,; may trigger competitor response,
- **What to measure:** conversion rate,; unit sales,
- **Guardrails:** margin floor protection,; inventory depletion limits,

A practical pricing architecture

- **Decision flow:** gather product, demand, inventory, and market signals,; forecast demand under candidate price scenarios,
- **Why architecture matters:** Pricing is not just a model output.

Common failure modes

- forecasting ignores stockouts and mistakes absence of sales for lack of demand,
- elasticity estimates are biased by historical policy,
- optimization recommends prices that violate margin or contractual constraints,
- promotions are evaluated on revenue alone while margin collapses,
- competitor signals are stale or incomparable,
- different channels display conflicting prices due to synchronization delays.

Hands-on lab: forecast plus constrained optimization

- **Lab scenario:** Students are given historical SKU sales, prices, promotion flags, inventory levels, and seasonality indicators.
- **Lab tasks:** Build a baseline demand forecast for a selected product or category.; Estimate how demand changes across a small set of candidate prices.
- **Expected learning outcome:** By the end of the lab, students should be able to explain why pricing requires a pipeline of assumptions, constraints, and measurement rather than a single predictive model.

Managerial implications

- clear pricing objectives,
- data quality around cost, stock, and catalog attributes,
- explicit separation between descriptive analytics and prescriptive decisions,
- and controlled experimentation before scaling automated pricing.